

A. Multiple Choice Select on the right answer with a circle.

1. **Cloud Computing:** A model for enabling on-demand network access to a shared pool of configurable computing resources

A) True

B) False

2. **Horizontal Scale-Out:** Adding more resources (like RAM or CPU) to a single computation unit to increase power.

A) True

B) False

Vertical Scale-Up: Adding more resources (like RAM or CPU) ...



3. **Vertical Scale-Up:** Adding discrete resources (like multiple machines) and making them behave as a converged unit.

A) True

B) False

Horizontal Scale-Out: Adding discrete resources (like multiple machines) ...



4. For HPC, Horizontal Scale-Out is better and less costly than Vertical Scale-Up

A) True

B) False

5. **CAPEX model:** Buy the dedicated hardware and pay for it over a period of time

A) True

B) False

6. **OPEX model:** Use shared cloud infrastructure and pay only as you use it

A) True

B) False

7. For moving to cloud, we move from OPEX to CAPEX model

A) True

B) False

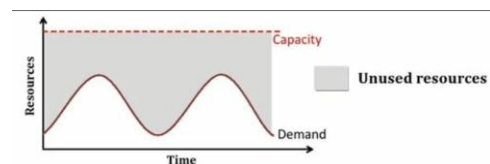
CAPEX to OPEX model

8. **Under-provisioning:** Allocating more resources than required, leading to wasted costs

A) True

B) False

Over-provisioning: Allocating more resources than required, leading to wasted costs

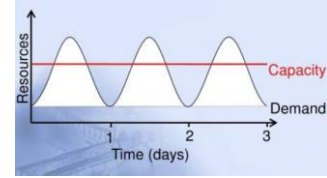


9. **Over-provisioning:** Allocating fewer resources than required, leading to poor service performance and loss of customers.

A) True

B) False

Under-provisioning: Allocating fewer resources than required, leading to poor ...



10. Elasticity helps match resources with demand

A) True

B) False

11. **Dynamic provisioning:** Automatically allocates/de-allocates resources based on usage

A) True

B) False

12. **Public Cloud:** The cloud infrastructure is provisioned for exclusive use by a single organization

A) True

B) False

Private Cloud: The cloud infrastructure is provisioned for exclusive use by a single organization

13. **Hybrid Cloud:** A composition of two or more distinct cloud infrastructures (private, community, or public)

A) True

B) False

14. **Cloud Bursting:** When an application in a private cloud moves to a public cloud to handle a spike in demand.

A) True

B) False

15. **Multi-cloud:** When multiple cloud vendors provide services together in a heterogeneous architecture.

A) True

B) False

16. **Type 1 Hypervisor (Bare Metal):** Runs directly on the guest operating system (OS)

A) True

B) False

Runs directly on the Hardware

17. A virtual machine (VM) is a software implementation of a computer that executes programs.

A) True

B) False

18. **Hypervisor:** Provides a uniform abstraction of the underlying physical machine.

A) True

B) False

19. **Type 2 Hypervisor (Hosted):** The hypervisor runs on a guest OS, like KVM

A) True

B) False

20. RESTful Web services use a uniform and predefined set of stateless operations.

A) True

B) False

21. **Cloud Broker:** A party that provides connectivity and transport of cloud services from providers to consumers.

A) True

B) False

Cloud Carrier: A party that provides connectivity and transport of cloud services from providers to consumers.

22. **Cloud Consumer:** A person or organization that maintains a business relationship with, and uses service from, Cloud Providers.

A) True

B) False

23. **Cloud Carrier:** An entity that manages the use and delivery of services and negotiates relationships between providers and consumers

A) True

B) False

Cloud Broker: A party that provides connectivity and transport of cloud services from providers to consumers.

24. **Cloud Auditor:** A party that can conduct independent assessments of security, performance, and information system operations.

A) True

B) False

25. **Cloud Provider:** A person, organization, or entity responsible for making a service available to interested parties.

A) True

B) False

26. **PaaS Consumer:** These are typically system administrators who manage physical hardware.

A) True

B) False

IaaS Consumer: These are typically system administrators who manage physical hardware.

27. **SaaS Consumer:** Typically end users who directly use software applications or software administrators who configure applications for end-users.

A) True

B) False

28. **IaaS Consumers:** These are typically application developers who design, test, and deploy software

A) True

B) False

PaaS Consumer: These are typically application developers who design, test, and deploy software

29. **WSDL (Web Service Description Language):** Is used by the client to understand how exactly to invoke a web service.

A) True

B) False

30. **HTTP Methods (GET):** Is used to create a new resource using the instructions in the request body.

A) True

B) False

HTTP Methods (POST): Is used to create a new resource using the instructions in the request body.

31. **Mashup:** A phenomenon to create new services by combining data or functionality from two or more resources.

A) True

B) False

32. **Host machine:** The virtual machine (VM) that runs on top of the hardware

A) True

B) False

Host machine is hardware; Guest machine is the virtual machine that runs on top of the

33. **Nested Virtualization:** The process of running a virtual machine within another virtual machine.

A) True

B) False

34. **Hardware-assisted virtualization:** Provides complete isolation with minimal performance overhead.

A) True

B) False

35. **Full virtualization:** Requires the guest operating systems to be adapted or modified.

A) True

B) False

Paravirtualization: Requires the guest operating systems to be adapted or modified.

36. **Amazon Machine Image (AMI):** A template for the instance (for example, an operating system, an application server, and applications)

A) True

B) False

37. **Storage Area Network (SAN):** A dedicated network that provides access to consolidated, block-level data storage.

A) True

B) False

38. **SAN Benefit:** Allows servers to boot from the SAN itself, enabling quick replacement of faulty servers

A) True

B) False

39. **NoSQL databases:** Database systems that work only with tabular relations like traditional SQL.

A) True

B) False

“Not Only SQL” i.e. database systems that works on other than the tabular relations used in relational databases.

40. **CAP Theorem:** It is impossible for a distributed system to simultaneously provide Consistency, Availability, and Partition tolerance.

A) True

B) False

41. **Eucalyptus:** An open-source platform for private clouds that is compatible with Amazon EC2 APIs.

A) True

B) False

42. **Content Delivery Network (CDN):** Used to increase latency by storing data far from the end users.

A) True

B) False

CDN is a geographically distributed network of proxy servers and their data centers, to provide high availability and performance

43. **Orchestration:** The automated arrangement, coordination, and management of complex workflows and resource lifecycles.

A) True

B) False

44. **Google File System (GFS):** A scalable distributed file system for large distributed data-intensive applications.

A) True

B) False

45. **Google App Engine (GAE):** A popular example of Infrastructure as a Service (IaaS).

- A) True
- B) False

GAE is a PaaS that allows developers to deploy applications without managing servers, with automatic scaling and maintenance.

46. For Containers, programs inside a container can see all files and devices on the host system.

- A) True
- B) False

For Containers, programs running inside of a container can only see the container's contents and devices assigned to the container.

47. Container is a running process interacting with its own private filesystem provided by a Container image.

- A) True
 - B) False
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48. Docker is a platform for developers and sysadmins to build, run, and share applications with containers.

- A) True
 - B) False
-

49. All containers are considered heavier and consume more resources compared to traditional Virtual Machines.

- A) True
- B) False

All containers are run by a single operating system kernel and therefore use fewer resources than VMs.

50. **DockerHub:** A popular example of Software as a Service (SaaS).

- A) True
- B) False

DockerHub is a Container as a Service (CaaS) repository used to store and share Docker images.

51. Docker can build images automatically by reading the instructions from a Dockerfile.

- A) True
 - B) False
-

52. **Dockerfile:** A text document that contains all the commands a user could call on the command line to assemble an image.

- A) True
 - B) False
-

53. **Container as a Service (CaaS):** A service model that allows users to manage and deploy containers.

- A) True
 - B) False
-

54. CaaS do not include development tools.

A) True

B) False

55. PaaS is for application development and deployment, while CaaS is for managing container environments.

A) True

B) False

56. **Distributed Version Control (DVC):** Single server holds the code base.

A) True

B) False

Distributed Version Control (DVC): Every developer has a full copy of the project history on their local machine.

57. **Centralized Version Control (CVC):** Every developer has a full copy of the project history on their local machine.

A) True

B) False

Centralized Version Control (CVC): Single server holds the code base.

58. **Version Control System (VCS):** A system that records changes to a file or set of files over time so that you can recall specific versions later.

A) True

B) False

59. **GitHub:** A web-based hosting service for version control using Git.

A) True

B) False

60. Cons of Centralized Version Control (CVC): Single point of failure.

A) True

B) False

61. Advantages of Distributed Version Control (DVC): Many operations, no single point of failure

A) True

B) False

62. Git is a centralized version-control system.

A) True

B) False

Git is a distributed version-control system.

63. **Git-LFS (Large File Storage)**: replaces large files such as audio samples, videos, datasets, and graphics with text pointers inside Git.

A) True

B) False

64. **Distributed File System (DFS)**: The primary goal is to move data to workers to save network bandwidth.

A) True

Distributed File System (DFS): The primary goal is to move workers to data to save network bandwidth.

B) False

65. The 5 characteristics of Big Data (5Vs) include: Volume, Variety, Velocity, Veracity, and Value.

A) True

B) False

66. GFS and HDFS are distributed computing environment to process "Big Data".

A) True

Google File System (GFS)

B) False

Hadoop Distributed File System (HDFS)

67. GFS and HDFS provide high aggregate performance to a large number of clients.

A) True

B) False

68. **Coarse-grained sharing**: This approach improves resource utilization by multiplexing small tasks across all available resources.

A) True

Fine-grained sharing: This approach improves resource utilization by multiplexing small tasks across all available resources.

B) False

69. The main components and ecosystem of Big Data: Data analytics, technologies, and visualization.

A) True

B) False

70. **Hadoop**: An open-source framework designed for the distributed processing of large data sets across clusters of computers using simple programming models.

A) True

B) False

71. The Datanodes manages metadata, file-to-block mapping, and coordinates access to data in HDFS.

A) True

The Namenode (master) manages metadata, file-to-block mapping, and coordinates access to data in HDFS.

B) False

72. Namenode store data blocks and handle replication, integrity, and storage management.

A) True

Datanodes (slave) store data blocks and handle replication, integrity, and storage management.

B) False

73. MapReduce is a programming model and an implementation for processing and generating large data sets.

A) True

B) False

74. MapReduce will not work for inter-process communication and data sharing requirements.

A) True

B) False

75. **MapReduce:** The 'Reduce' function is used to extract data of interest from each individual record.

A) True

Map extracts data, and Reduce aggregates it to generate the final output.

B) False

76. **MapReduce:** The 'Map' function is used to aggregate intermediate results and generate the final output.

A) True

Map extracts data, and Reduce aggregates it to generate the final output.

B) False

77. CDN improves performance by distributing content from locations near the end-user.

A) True

B) False

78. **Over-provisioning:** Allocating more resources than required, leading to underutilization and wasted cost.

A) True

B) False

79. Cloud Computing requires high upfront cost to buy dedicated hardware.

A) True

B) False

80. Grid Computing commonly uses commodity nodes/resources to provide scalable services.

A) True

Cloud Computing uses commodity nodes

B) False

Grid Computing uses custom nodes.

81. Cloud Computing commonly uses custom or heterogeneous nodes shared across different organizations.

A) True

B) False

Cloud Computing uses commodity nodes

Grid Computing uses custom nodes.

82. **Cloud Architecture:** Fabric layer contains raw hardware level resources such as compute, storage, and network resources.

A) True

B) False

83. **Google Content Management:** It can be used for web information, marketing campaigns, or social media.

A) True

B) False

84. **Service Analysis:** Identification and contextualization of market requirements and service reuse.

A) True

B) False

85. **Service Design:** It includes detailed model of the service and refining the service idea for implementation.

A) True

B) False

86. **Service Implementation:** It means registering the service in marketplaces and repositories.

A) True

B) False

Service Implementation: either software with technical service characteristics or executable non-technical service.

87. **Service Publishing:** It includes access rights, costs, pricing models, and SLAs.

A) True

B) False

88. **Service Retirement:** It is monitored for contract management, SLA compliance, billing, and service performance management.

A) True

B) False

Service Operation: It is monitored for contract management, SLA compliance, billing, and service performance management.

89. **Service Operation:** It happens when the service reaches the end of economic or technical competitiveness.

A) True

B) False

Service Retirement: It happens when the service reaches the end of economic or technical competitiveness.

90. Cloud Service life cycle includes: Requirements, Discovery, Negotiation, Composition, and Consumption.

A) True

B) False

91. Multiple Cloud Systems serially means using more than one cloud system at the same time.

A) True

Multiple Cloud Systems serially means using more than one cloud system **one at a time**.

B) False

92. **Case 1A:** Private cloud within a single administrative domain and trust boundary.

A) True

B) False

93. **Case 1B:** Commercial public cloud within a single administrative domain and inside the client trust boundary.

A) True

Case 1B: Commercial public cloud within a single administrative domain and **outside** the client trust boundary.

B) False

94. **Case 2A:** Federated cloud of two or more administrative cloud domains in a common trust boundary.

A) True

B) False

95. **Case 2B:** Hybrid cloud applications cross a private-public trust boundary.

A) True

B) False

96. NetBeans IDE supports Web services standards such as JAX-WS, JAX-RS, and JAXB.

A) True

B) False

97. **NetBeans SOAP-based Web Services:** can be converted to RESTful service resources using the action available in the web service node.

A) True

B) False

98. REST HTTP PUT is used to add new data.

A) True

REST HTTP **POST** is used to add new data.

B) False

99. REST HTTP POST is used to update data that already exists.

A) True

REST HTTP **PUT** is used to update data that already exists.

B) False

100. RESTful Web Services can provide CRUD functionality using methods such as Read, Write, Update, and Remove.

A) True

RESTful Web Services can provide CRUD functionality using methods such as **GET, POST, PUT, and DELETE**.

B) False

101. NetBeans Web Service Clients can generate code for invoking both RESTful and SOAP-based Web Services.

A) True

B) False

102. NetBeans Web Service Clients can create server-side mash-up applications and add services from WSDL or WADL files.

A) True

B) False

103. NetBeans Web Service Manager can access popular RESTful Web APIs provided by Google, Facebook, Yahoo, Flickr, Amazon, Twitter and many more.

A) True

B) False

104. **NetBeans Web Service Interoperability Technologies (WSIT)**: help Web Services work with different platforms and systems.

A) True

B) False

105. NetBeans supports Apache Axis, while Eclipse supports GlassFish Server.

A) True

NetBeans supports GlassFish, while Eclipse supports Apache Axis Server.

B) False

106. **Eclipse Web Services Tools**: WST is used for Java Web Services development.

A) True

WST Web Services component contains tools for Web services development which is **not Java specific**.

B) False

107. **Eclipse Web Services Tools**: J2EE Standard Tools (JST) contains tools for developing and interacting with Java Web services.

A) True

B) False

108. Apache JMeter is used to load test functional behavior and measure performance.

A) True

B) False

109. Apache JMeter can simulate a heavy load on a server, group of servers, network, or object.

A) True

B) False

110. SoapUI is used to create, manage, and execute tests on REST, SOAP, and GraphQL APIs.

A) True

B) False

111. SoapUI API Testing Tool supports scriptless functional testing, security testing, load testing, API mocking, and protocols such as SOAP/WSDL, REST, and JMS.

A) True

B) False

112. **Inter-operational analysis:** inspects code within a single operation.

A) True

Intra-operational analysis: inspects code within a single operation.

B) False

113. **Intra-operational analysis:** inspects relationships between operations and procedures.

A) True

Inter-operational analysis: inspects relationships between operations and procedures.

B) False

114. **Web Service Slicing:** a technique that captures a functional subset of a large-scale web service using an Interface slice

A) True

B) False

115. A subset service provides full of the original service's functionality.

A) True

A subset service provides only some of the original service's functionality.

B) False

116. Interface slice captured as a subset of the original interface's provides access to an Interoperable slice

A) True

B) False

117. An interoperable slice is a portion of the web service code that can be accessed as a subset service from a remote client.

A) True

B) False

118. **Unit WSDL (UWSDL):** captures changes in the WSDL.

A) True

B) False

Difference WSDL (DWSDL): captures changes in the WSDL.

119. **Difference WSDL (DWSDL):** captures changes in the Web Service code.

A) True

B) False

Unit WSDL (UWSDL): captures changes in the Web Service code.

120. Inter-operational change analysis captures changes that depend on changed operations and procedures.

A) True

B) False

121. **Parameter WSDL (PWSDL):** captures code flow changes at the Web Service code level.

A) True

B) False

122. PRTWS uses intra-operational change analysis, while ORTWS uses inter-operational change analysis.

A) True

B) False

ORTWS uses intra-operational change analysis, while **PRTWS** uses inter-operational change analysis.

123. **Automated Web Service Change Management (AWSCM):** a tool for change analysis based regression testing of web service.

A) True

B) False

124. **White box analysis:** The internals of the Web Service code are known.

A) True

B) False

125. **Black box analysis:** The internals of the Web Service code are unknown.

A) True

B) False

126. **Service Change Classifier:** change labels to operations that extracts WSDL slice.

A) True

B) False

127. Service Evolution Metrics are based on number of operations, WSDL lines, parameters, messages, and operation code lines.

A) True

B) False

128. Subset regression testing helps to maintain the QoS and SLA.

A) True

B) False

129. **Quality of Service (QoS):** A contract between service provider and consumer.

A) True

B) False

Quality of Service (QoS): Service performance measurement in cloud computing service.

130. **Service Level Agreement (SLA):** Service performance measurement in cloud computing service.

A) True

B) False

Service Level Agreement (SLA): A contract between service provider and consumer.

131. Throughput, availability, and jitter are examples of QoS measurements.

A) True

B) False

132. SLA includes quality, availability, and responsibilities.

A) True

B) False

133. **Web Service Level Agreement (WSLA):** SLA for Web service monitoring.

A) True

B) False

134. SLA contract includes QoS parameters, SLA negotiation, monitoring, violation detection, and enforcement.

A) True

B) False

135. Monitoring is useful for preventing failures, verifying service invocation conditions, and triggering recovery actions when needed.

A) True

B) False

136. WS-I promotes interoperability across platforms, operating systems, and programming languages

A) True

B) False

137. WS-I helps define protocols for interoperable exchange of messages between Web services.

A) True

B) False

138. Regression testing has no relation to maintaining QoS and SLA.

A) True

Regression testing helps maintain QoS and SLA after service changes

B) False

139. Reducing regression testing effort can reduce the effort required to monitor and maintain QoS.

A) True

B) False

140. Change analysis enables regression testing of Web services to manage QoS and enforce SLA requirements.

A) True

B) False

141. Hadoop is used for real-time processing, while Storm is used for batch processing.

A) True

Storm is used for real-time processing, while Hadoop is used for batch processing.

B) False

142. **Big Data Volume:** speed of generation and processing to meet the demands

A) True

Big Data Volume: Refers to the large size of data, such as terabytes and petabytes.

B) False

143. **Apache Storm:** A free and open-source distributed real-time computation system used to process unbounded streams of data.

A) True

B) False

144. **Storm Topology:** A topology is a graph of spouts and bolts connected with stream groupings.

A) True

B) False

145. **Storm Stream:** A stream is an unbounded sequence of tuples processed in parallel in a distributed fashion.

A) True

B) False

146. A MapReduce job runs forever, while a Storm topology finishes after processing data.

A) True

B) False

A topology is analogous to a MapReduce job, but **a topology runs forever or until you kill it.**

147. **Storm Spout:** A spout is a source of streams in a topology.

A) True

B) False

148. **Storm Bolt:** All processing in topologies is done in bolts.

A) True

B) False

149. **Apache Spark:** A unified analytics engine for large-scale data processing.

A) True

B) False

150. Apache Spark is only used for batch processing and cannot process streaming data.

A) True

B) False

Apache Spark can process **both** batch and streaming data.

151. Spark architecture depends on two abstractions: RDD and DAG.

A) True

B) False

152. **Resilient Distributed Datasets (RDD):** A distributed memory abstraction used to perform in-memory computations on large clusters.

A) True

B) False

153. Keeping data in memory can improve performance in Spark.

A) True

B) False

154. **Wide dependency:** Each partition of the parent RDD is used by at most one partition of the child RDD.

A) True

B) False

Narrow dependency: Each partition of the parent RDD is used by at most one partition of the child RDD.

155. **Narrow dependency:** where multiple child partitions may depend on it.

A) True

B) False

Wide dependency: where multiple child partitions may depend on it.

156. **Direct Acyclic Graph (DAG):** is used to perform a sequence of computations in Spark.

A) True

B) False

157. When an action such as count or save runs on an RDD, Spark builds a DAG of stages to execute.

A) True

B) False

158. Narrow dependency requires shuffle operations between stages.

A) True

B) False

Wide dependency requires shuffle operations between stages.

159. DataFrame is a collection of data organized into named columns.

A) True

B) False

160. FP stands for Frequent Pattern in a dataset of transactions.

A) True

B) False

161. **Sharding:** Dividing the database into parts and storing them on different computers.

A) True

B) False

162. **Availability of a Service:** Multiple datacenters always remove all single point of failure risks.

A) True

B) False

A single cloud company can still become a single point of failure, so the vendor needs business-continuity strategy agreements.

163. **DDoS attack:** A multitude of compromised systems attack a single target, causing denial of service for users of the targeted system.

A) True

B) False

164. **Data Lock-In:** Customers cannot easily extract their data and programs from one cloud site to run on another.

A) True

B) False

165. To reduce Data Lock-In risk, storing all data with only one cloud provider is the best solution to avoid losing access to data.

A) True

B) False

To reduce Data Lock-In risk, **standardizing APIs** and storing data on **multiple cloud providers** can reduce the risk of losing access to data if a single cloud company fails.

166. **Data Confidentiality and Auditability:** Encrypted storage, VLANs, firewalls, and auditability can help protect sensitive data in the cloud.

A) True

B) False

167. **Data Transfer Bottlenecks:** Shipping physical disks can help overcome the high cost and long time of Internet data transfers.

A) True

B) False

168. Sharing I/O efficiently in cloud computing is not easy, and this can cause performance unpredictability.

A) True

B) False

169. **Scalable Storage:** It requires high up-front cost and fixed storage capacity.

A) True

Scalable storage aims for **no up-front cost**, **short-term usage**, and **on-demand capacity**.

B) False

170. Bugs in Large-Scale Distributed Systems are not easily reproduced in smaller configurations, so debugging often occurs at datacenters.

A) True

B) False

171. A customer's bad behavior can affect the reputation of the cloud, so reputation-guarding services are needed.

A) True

B) False

172. **Availability of Service:** Cloud elasticity can help defend against DDoS attacks by scaling resources during sudden traffic increases.

A) True

B) False

173. Service providers may access cloud data, so users can encrypt data before storing it in the cloud to prevent unauthorized reading.

A) True

B) False

174. Physical control on the computers in public cloud is more secure than in private cloud.

A) True

Physical control on the computers in private cloud is more secure than in public cloud.

B) False

175. **NIST Guidelines:** Continuous monitoring helps maintain awareness of security controls, vulnerabilities, and threats.

A) True

B) False

176. GDPR is a regulation in European Union law on data protection and privacy, and its primary aim is to give individuals control over their personal data.

A) True

B) False

177. CDN providers may use user analytics and tracking data for behavioral targeting, which can cause privacy intrusion.

A) True

B) False

178. Same-Origin Policy permits scripts if pages have the same origin: URI scheme, host name, and port number.

A) True

B) False

179. **CDN Security and Privacy:** JavaScript served by CDN can be targeted for malicious code injection, and Subresource Integrity helps verify the content using a hash reference.

A) True

B) False

180. **Edge Computing:** Computation and data storage are placed far from the user location to increase response time and bandwidth usage.

A) True

B) False

Edge Computing places computation and data storage **closer to the user** to **improve response time** and **save bandwidth**.

181. **Ubiquitous Computing:** Computing can appear anytime and everywhere, using devices such as tabs, pads, and boards.

A) True

B) False

182. **Cloudlet:** A large-scale central cloud datacenter located far from users at the core of the Internet.

A) True

B) False

Cloudlet: A mobility-enhanced small-scale cloud datacenter located at the edge of the Internet.

183. Fog Computing is defined by the exclusion of cloud, whereas Edge Computing works with the cloud.

A) True

B) False

Fog Computing **works with the cloud**, whereas edge computing is defined by the **exclusion of cloud**.

184. **IoT:** A network of physical objects embedded with sensors, software, and technologies to exchange data over the Internet.

A) True

B) False

185. **Virtual Event:** An online event where people interact in a virtual environment instead of a physical meeting.

A) True

B) False

186. **Hybrid Event:** Combines a physical in-person event with a virtual online component.

A) True

B) False
